The Verification Gap: Checking the Reasoning in AI-Assisted Academic Writing

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Abstract

Generative AI tools now help write a growing share of academic papers. Checking that their citations exist has received attention, but a paper whose contribution is an argument can cite only real sources and still reason badly. Reporting guidelines, such as those collected by the EQUATOR (Enhancing the QUAlity and Transparency Of health Research) Network, help authors report a study completely; whether the reasoning from evidence to claim holds is a separate question, and it is most exposed in papers that have only reasoning to offer. This commentary proposes a claim register: each claim is recorded with what it rests on, each kind of grounds gets its own check, the step from grounds to claim is checked for relevance and sufficiency, and the wording of each claim is tied to how well it is supported. Each part is already in use, in safety engineering, scholarly data publishing, educational measurement and the calibrated reporting of certainty; none of the approaches compared here combines them for academic papers. Publishing a register can make a paper's reasoning visible and traceable. Whether it helps authors and reviewers find weak steps has not been tested. The paper applies the register to itself and reports what it shows.

Keywords: academic writing, generative AI, verification, argumentation, reporting guidelines, non-empirical research

Key Points:

- Generative AI tools can produce fluent papers whose citations are real and whose reasoning is still weak.
- Reporting guidelines help authors report a study completely; they do not ask whether each step of reasoning holds.
- A claim register records what each claim rests on and checks each step by its kind, with wording tied to support.
- Each part of the register is used elsewhere; none of the approaches compared combines them for academic papers.
- Publishing a register makes reasoning traceable; whether it helps reviewers find weak steps is untested.

1 Introduction

Writing assistants based on large language models (LLMs) have spread quickly: up to 22% of computer science papers now show signs of LLM editing, and such editing can be detected in every major discipline (Liang et al., 2025). In this paper "generative AI tools", or "AI" for short, means these assistants. When a human author falls short, the text tends to show it, in awkward phrasing or gaps in coverage. An AI-assisted draft can read as polished and complete while it contains errors that only checking would reveal, such as a citation to a paper that does not exist.

Citation hallucination is the most discussed of these errors. Mugaanyi et al. (2024) found that 62–89% of the digital object identifiers (DOIs) that LLMs generated for natural-science and humanities topics were fabricated, and Walters and Wilder (2023) found fabricated citations among the references of generated literature reviews, with fewer from the newer model they tested (18% for OpenAI's GPT-4 against 55% for GPT-3.5). A fabricated citation can be caught by looking the work up.

A second error is harder to see. I call it *confidence inflation*: wording that expresses more certainty than the evidence supports. The evidence indicates that LLMs verbalise high confidence even on questions they answer wrongly (Xiong et al., 2024; Zhou et al., 2024). Whether this carries over into academic prose is unsettled: in one corpus of applied-linguistics abstracts, ChatGPT used boosters such as "shows" at twice the human rate, but also more hedges (El-Dakhs et al., 2025). The stakes are epistemic, because words such as "demonstrates" and "suggests" are meant to tell a reader how much weight a claim can bear.

A citation check, as I use the term, asks whether a cited work exists and says what is attributed to it. It catches the first error. It does not reach the second, or a step from real evidence to a claim the evidence does not carry, so a fluent paper can pass it with its reasoning unchecked.

Detecting AI-written text does not fill that gap. Detection asks who wrote a text; verification asks whether it holds. Detectors have also been fragile in tests: their accuracy varies with generation settings (Dugan et al., 2024), and paraphrase can collapse it

(Krishna et al., 2023).

2 What Reporting Guidelines Check

The most developed infrastructure for checking academic writing is the EQUATOR Network's collection of about 700 reporting guidelines for health research (EQUATOR Network, n.d.). Guidelines such as CONSORT for randomised trials and PRISMA for systematic reviews (Page et al., 2021) give authors a checklist to complete while they prepare a manuscript, and the GoodReports tool picks the checklist that fits a study type (Struthers et al., 2021).

What these guidelines check is reporting. Their developers define a reporting guideline as "a checklist, flow diagram, or explicit text to guide authors in reporting a specific type of research" (Moher et al., 2010), and the effect that has been measured is on completeness: journals endorsing CONSORT reported more completely on 25 of 27 checklist items, significantly so for five (Turner et al., 2012). Incomplete reporting is a recognised source of research waste (Glasziou et al., 2014), so this is a real service. It answers a different question from whether the reasoning holds, and that second question applies to any paper, empirical or not.

The second question matters most where a paper has little else to check. Gregor (2006) distinguishes five types of theory in information systems research, among them analytic theories (Type I) and theories for design and action (Type V). When a paper of either type proposes a new theory, no earlier source states it, because the proposal is the contribution. I call papers whose main contribution is reasoning *non-empirical*: theoretical, conceptual, design-science and perspective papers are examples, and the list is not a classification.

In such a paper the reasoning carries the contribution. Toulmin (2003) places the step from evidence to claim in the warrant, and Whetten (1989, p. 491) calls the reasoning of a theory "the theoretical glue that welds the model together". A check that stops at citations may therefore leave a non-empirical paper's contribution itself unchecked.

Guidance for papers without new data exists, but it is scattered. Conceptual papers

lack "commonly accepted templates" (Jaakkola, 2020, p. 18), although Jaakkola proposes four. Design-science researchers have reported no agreed guidance on presenting their work (Gregor and Hevner, 2013; Dinter and Krawatzeck, 2015), and validity in that field "is neither well-understood nor consistently established" (Larsen et al., 2025). Software engineering's community standards ask methodological papers for "clear, valid arguments" but give no procedure for checking them (Ralph et al., 2020). To look further, I searched the EQUATOR library (709 titles) and four bibliographic sources (Europe PMC, OpenAlex, Crossref and arXiv) on 16 October 2026, with 28 queries and 701 titles screened by one screener, for a consensus-based checklist that checks the reasoning of non-empirical papers step by step: whether each step from evidence to claim holds. I found none. The search did not cover Scopus, Web of Science or humanities indexes, and a second screener has not yet checked it.[1]

3 Related Approaches

Several practices already record or check parts of an argument (Table 1).

Recent tools for AI-assisted writing go beyond checking that references exist: a linter that also checks causal language against the study design (Samsonau, 2026), a workflow that returns unsupported claims to the author (Chen et al., 2026), a provenance discipline that samples whether the evidence suffices (Zhou and Yu, 2026), and a checklist in development for cited assertions in narrative reviews (VANRA Development Group, 2026). As their authors describe them, each asks whether evidence exists and supports a claim; none describes a check of whether the step from grounds to claim holds for its readers.

The appraisal tools come closer to the reasoning. JBI's consensus-based checklist for appraising expert opinion asks "Does the opinion demonstrate a logically defended argument to support the conclusions drawn?" and refers reviewers to Toulmin's model (McArthur et al., 2025; JBI, 2026). SANRA, a scale for narrative reviews, scores scientific reasoning as one of its six items, and its authors call it "not a reporting guideline"

[1]Protocol, queries and screened candidates are in the repository (see Code and Data Availability), so the search can be repeated.

Table 1: Approaches that record or check arguments. The last column is this paper's reading of each source.

Approach	**Scope**	**What it checks**	**Left to judgement**
Reporting guidelines (Moher et al., 2010)	One type of empirical study	That the items of the study are reported	Whether the conclusions follow
Appraisal tools (JBI, 2026; Baethge et al., 2019)	Expert opinion (JBI); narrative reviews (SANRA)	One question on the argument as a whole (JBI); six items scored 0–2, one of them scientific reasoning (SANRA)	How each step is judged
AI-writing checkers (Samsonau, 2026; Chen et al., 2026; Zhou and Yu, 2026)	AI-assisted manuscripts	That cited evidence exists and supports a claim; causal wording against design	Whether a step holds for readers
Assurance cases (Kelly and Weaver, 2004; Bloomfield and Netkachova, 2014)	Safety of a system	Claims linked to evidence through stated arguments and assumptions	Whether the case convinces
Micro- and nanopublications (Clark et al., 2014; Groth et al., 2010)	Statements in (mainly biomedical) publications	Records claims with attribution and support; no checking procedure	Whether the support holds
Argument-based validation (Kane, 2013)	Interpretation and use of test scores	Coherence, completeness and plausibility of the argument's inferences	How plausibility is weighed
Argument quality (Wachsmuth et al., 2017)	Arguments in natural language	Acceptability, relevance and sufficiency of premises	Not a procedure for papers
Calibrated language (Kent, 1964; Mastrandrea et al., 2010; Santesso et al., 2020)	Estimates, assessments, evidence summaries	Wording fixed per level of confidence	How readers read the words

(Baethge et al., 2019); reporting items for concept analysis have only recently been collected (Nashwan et al., 2026). Both judge an argument as a whole.

Two practices record the structure of an argument explicitly. In safety engineering an assurance case, or safety case, is "a documented body of evidence that provides a convincing and valid argument that a system is adequately safe" for its use; its authors "make an explicit set of claims", link them to the evidence through arguments, and "make clear the assumptions and judgements underlying the arguments" (Bishop and Bloomfield, 1998). The Goal Structuring Notation breaks claims into sub-claims until each can be supported by evidence (Kelly and Weaver, 2004), and the claims–arguments–evidence approach defines its arguments as "Toulmin's warrants" and treats claims asserted without justification as assumptions (Bloomfield and Netkachova, 2014). In scholarly data publishing, nanopublications express a core scientific statement with its context and provenance (Groth et al., 2010), and micropublications record a statement with its attribution and, at their fullest, with the evidence and challenges brought for and against it; their model "is grounded in Toulmin-Verheij" (Clark et al., 2014). Both make claims and their support machine-readable. Neither describes a procedure for checking a step.

Educational measurement has an example of checking an argument step by step. In Kane's argument-based approach, the claims made from test scores are laid out as an argument that specifies the inferences and assumptions they need, and validation evaluates the coherence and completeness of that argument and the plausibility of its inferences and assumptions (Kane, 2013). The object is the use of a test, not a paper.

Research on argument quality supplies criteria for the step itself. Wachsmuth et al. (2017) distinguish logical, rhetorical and dialectical quality, and define a cogent argument as one with "individually acceptable premises that are relevant to the argument's conclusion and, together, sufficient to draw the conclusion". Walton et al. (2008) collect argumentation schemes, which their publisher describes as "forms of argument that capture stereotypical patterns of human reasoning".

Each part therefore has users. Claims are recorded with their grounds in assurance cases and micropublications; an argument is checked inference by inference in argument-

based validation, which draws on Toulmin's model; argument-quality research supplies criteria for a single step; and calibrated language ties wording to the strength of support. None of the approaches compared here combines these parts for an academic paper, and that combination is what the next section proposes.

4 A Register of Claims

The proposal borrows its vocabulary from Toulmin and Whetten. Box 1 gives the terms; the framework's glossary defines them from their sources.[2]

Box 1. Terms used in this paper.

Term	Meaning
Claim	Any assertion put forward for acceptance.
Grounds	What a claim rests on: a source, other claims, a declared starting point, or the author's own work. An extension of Toulmin's *data*.
Warrant	The step from grounds to claim.
Qualifier	The hedge that states how strong the claim is.
Rebuttal	Circumstances in which the step from grounds to claim no longer holds.
Confidence tier	The rating (Established, Supported, Emerging, Speculative) that decides which qualifiers a claim may carry.
Priority	How much the paper depends on a claim (P0 to P2).
Declared starting point	An assumption, definition or value the paper adopts without arguing for it.
Contribution	What the paper adds; checked once, as a whole.
Verification	Checking each claim against its grounds and warrant.
Validation	Checking that the contribution serves its readers and purpose.

Toulmin modelled his layout of arguments on jurisprudence, calling logic "generalised jurisprudence" (Toulmin, 2003, p. 7). In his account any assertion is a claim; its *data* are "the facts we appeal to as a foundation for the claim"; its warrant is one of the "general, hypothetical statements, which can act as bridges" from data to claim; a *qualifier* indicates "the strength conferred by the warrant on this step"; and a *rebuttal* names "circumstances in which the general authority of the warrant would have to be set aside" (Toulmin, 2003, pp. 90–94). Toulmin later wrote that he had not had in mind an analytical model like the one that came to be called "the Toulmin model"; his concern "was with twentieth-century epistemology, not informal logic" (Toulmin, 2003, p. vii). Using his layout as a checklist is my adaptation, and it has to be judged on its own merits.

[2]The glossary, the claim registry and the checking tools are public; see Code and Data Availability.

Others have put his elements to work. Micropublications build on his layout to record the claims of (mainly biomedical) publications (Clark et al., 2014); Kane's argument-based validation analyses the inferences from test scores to their uses with his model (Kane, 2013, pp. 11–12); and two studies of university students' essays scored argument quality element by element, with two raters each (Yang, 2022; Divya and Alamelu, 2026). The second study adapted the first one's rubric. None of these uses checks a research paper's reasoning, so that application is a further step.

Checks by kind of grounds. In the register, each claim is recorded with its grounds item by item, and each kind of grounds gets its own check:

- a **source** is checked in two steps: the work exists, and it says what the claim attributes to it;
- a **declared starting point** is checked for being stated, with a note on whether the field accepts or contests it;
- the author's **own work**, such as a search or a trial, is checked for a method someone else could repeat;
- a step from **other claims**, or from several sources to a point none of them makes alone, gets the warrant check.

The warrant check. The warrant check asks two questions, taken from Wachsmuth and colleagues' definition of a cogent argument. *Relevance*: does each ground bear on this claim, so that the claim would be less credible without it? *Sufficiency*: do the grounds together justify the claim at the strength its qualifier states? A sufficiency check names what is missing, such as a case not covered, a search not run or a premise that rests on one study. The check also records the rebuttal, the circumstances in which the step would fail, and the strongest known objection with the answer to it. Where a step follows a common pattern, such as an argument from expert opinion or from the absence of search results, the questions proper to that pattern could be added; working them out from the argumentation-schemes literature is still to be done. The check has four outcomes: it

passes, it passes and someone other than the author confirms it, it leaves a named gap open, or it fails. A failed check means that the recorded grounds do not carry the claim as worded, not that the claim is false; rewording the claim to what the grounds carry is the usual response.

Tiers and wording. Fixed wording for levels of certainty has precedents. Kent (1964) argued for "a few key odds expressions" in intelligence estimates; the Intergovernmental Panel on Climate Change (IPCC) asks its authors to judge confidence from the "type, amount, quality, and consistency of evidence" and "the degree of agreement" (Mastrandrea et al., 2010); and the GRADE working group (Grading of Recommendations Assessment, Development and Evaluation) fixes wording per level of certainty and size of effect, from "results in" to "may result in" (Santesso et al., 2020). Fixed terms may not secure a shared reading: when readers assigned numbers to the probability terms in an IPCC report, their judgements deviated from the IPCC's guidelines even when the guidelines were at hand (Budescu et al., 2009). Table 2 adapts the practice for academic papers. A claim's tier is the lower of two ratings: how well its grounds are supported, and what its warrant check allows. An argued claim can therefore reach the Supported tier when its premises are supported and its check passes, which a rule that only counts sources does not allow. The table has not been tested with readers; it may help an author notice when a sentence claims more than its grounds allow.

Table 2: Confidence tiers and the qualifiers they permit. A claim takes the lower of the two ratings.

Tier	**Support for the grounds**	**Warrant check**	**Qualifiers**
Established	Three or more independent sources agree, at least two of them peer-reviewed; no known dispute	Passed and confirmed by someone other than the author	"shows", "confirms"
Supported	Two independent peer-reviewed sources agree	Passed	"indicates", "evidence suggests"
Emerging	One peer-reviewed source; two or more that are not; or pilot data	A named gap left open	"may", "preliminary evidence"
Speculative	No source reports the point, or a single one that is not peer-reviewed	Failed	"warrants investigation"

Priority. Every claim also carries a *priority*, the author's answer to the question of what fails if the claim is wrong. A P0 claim is one the paper fails without; P1 strengthens the argument; P2 gives context. Priority is guidance on how much checking a claim deserves. It is not a gate: a paper whose central claims stay at Emerging is reporting how far its evidence goes, and the register makes that visible.

The contribution. Whetten's criteria work at a different level. He set out what "a complete theory must contain": which factors (What), how they relate (How), why ("the theoretical glue that welds the model together"), and for whom, where and when it holds (Whetten, 1989, pp. 490–492). His seven reviewer questions, from "What's new?" to "Who cares?", answer "What constitutes a publishable theory paper?" (Whetten, 1989, p. 494). They judge a contribution as a whole, so they may serve as a check of a paper's central contribution, applied once. For a contribution that is a recommendation, they fit less well: How tends to restate the thesis, and the check needs a declared value and the conditions under which the recommendation applies.

These two levels map onto a distinction from systems engineering. Verification is "proof of compliance with requirements"; validation shows that a product "accomplishes the intended purpose in the intended environment" (NASA, 2016, §2.0). Checking each claim against its grounds and warrant is verification in this sense; asking Whetten's questions of the contribution is closer to validation. The mapping is mine. The register then works as a traceability record, of the kind systems engineers keep for requirements: each link, from the contribution to its claims, from a claim to its grounds and from grounds to their sources, is written down, so that someone else can follow and check it. The IPCC asks its authors for a "traceable account" of their evaluation of evidence for the same reason (Mastrandrea et al., 2010).

What passing means. A register that passes its checks means that the paper followed the method: each claim's grounds and warrant were recorded and checked against stated criteria. It does not mean that the claims are true. A check run by the author, or by the author's AI tools, can err in the author's favour, which is why only a check confirmed by

someone else can reach the Established tier.

The central argument of the proposal rests on these pieces. A check that rates an inference by counting its sources does not look at the step from grounds to claim, so it cannot tell a sound inference from an unsound one, and it may fail a sound inference for the wrong reason. Recording each claim's grounds item by item, and letting the kind of grounds decide the check, may avoid such false failures. The argument rests on design reasoning and on two trial registries kept by the framework's own tooling, and no such false failure has yet been documented in a paper written by someone else.

5 Limitations

Its parts have users elsewhere (Section 3), but the register itself has one so far: the author of this paper. Whether it works in other hands is untested. Its reviews came from five language models from four vendors, which read the first two drafts in sessions separate from the drafting; they are independent of the drafting but not of the method, and no substitute for human reviewers. Whether tools can carry part of the work is also open. Structured prompting was reported to improve LLM argument extraction (Gupta et al., 2024), and representing support and attack formally to improve LLM review aggregation (Sukpanichnant et al., 2024). Both are narrow tasks, so they suggest only that structure combined with LLMs may help with checking as well.

The register asks work of authors; this paper's has 36 entries. A shorter form, the warrant check's questions for a reviewer to ask of the central claims, may suit many papers better. Toulmin's and Whetten's frameworks come from Western academic traditions, and scholars in the humanities may find a claim-by-claim layout foreign to how their arguments work; their fit with other conventions needs to be assessed.

The checks are also partly circular: the author who wrote the rules applies them, and the warrant check is a judgement, recorded with its reasons so that someone else can disagree with it.

6 Recommendations

The value behind this proposal is that published reasoning should be checkable: readers and reviewers should be able to see what each claim rests on. A claim register, published with a paper whose contribution is an argument, makes that reasoning visible and traceable. I propose two steps, in this order. Authors and journals could publish registers as optional supplementary material, as this paper does, and trials should test whether reviewers who have a register find different weaknesses from reviewers who do not. A register should become part of verification infrastructure for non-empirical papers, of the kind EQUATOR offers for empirical health research, only if such trials find that it helps. A first implementation exists: templates for the register and the glossary, the checks for each kind of grounds, and tools that test a register against its manuscript (see Code and Data Availability).

The proposal applies to papers whose contribution is an argument, and to the interpretive parts of empirical papers. Other writing that asks readers to trust its reasoning, journalism among them, might use it too, but that is untested. Peer review already judges arguments, and a register would not replace it. It could show reviewers where a paper's weight rests, and it could help authors find weak steps before submission; whether it does either is the question the trials should answer.

Several further directions warrant investigation: a second screener and wider databases for the gap search, the questions proper to common patterns of argument, adapting the checks to other fields, and building them into writing tools so that they run while an author drafts.

Acknowledgements and use of AI

The author conceived the idea and developed the framework, starting in March 2026, and made the decisions on scope, structure and the framework's rules. Generative AI was used throughout, under the author's direction: Claude Opus 5.5 (Anthropic), through Claude Code, drafted and revised the prose, searched for and checked sources, drafted the

claim register and ran the framework's checking tools; the author directed the revisions and reviewed each draft. Five models from four vendors reviewed the first two drafts and the framework's glossary in sessions separate from the drafting: DeepSeek-V4-Pro, Gemini 3.1 Pro (preview), GPT-6 (OpenAI, through ChatGPT), and Claude Opus 5.5 and Claude Fable 5.1 (Anthropic); their reviews informed this version. The author takes full responsibility for the content. [**Verification record, to be filled from the tool runs of the submitted version, with their date.**]

Code and data availability

The framework described here (templates, glossary and checking tools) and this paper's claim register are public under a CC BY 4.0 licence at `https://github.com/ducroq/agent-ready-papers`. The version this paper describes is [**release, to be set at submission**], archived at [**Zenodo DOI, to be added when the release is archived**].

A Self-Application

This paper was written with the procedure it proposes. Its claim register, published with the paper,[3] records each claim's grounds, warrant, warrant check, qualifier, confidence tier and priority, plus the paper's declared starting points and its one contribution. Table 3 shows three entries: a claim resting on sources, a claim resting on the author's own search, and the contribution.

What the register shows about this paper. Of the six P0 entries, three reach the Supported tier and three stay at Emerging: the gap claim (S2-2), the central argument (S3-4) and the contribution (C1), which rests on both. Under the rule in Table 2 they stay at Emerging for reasons that are named and could be removed: the search had one screener (S2-2), and the trial registries were kept by one author's tooling (S3-4). A second screener and a register kept by someone else are the evidence this paper still lacks.

[3] `https://github.com/ducroq/agent-ready-papers/blob/master/papers/perspective/vv/claims/claim_registry.md`

Table 3: Three register entries: what each rests on, how it was checked, and how its priority and confidence tier were set.

	S1-1	**S2-2**	**C1 (contribution)**
Statement	AI citation hallucination is a distinct failure mode.	No consensus checklist was found for checking the reasoning of non-empirical papers step by step.	A published register makes reasoning traceable; registers should be trialled before they are adopted.
Says	descriptive	descriptive	normative
Grounds	two sources: Mugaanyi et al. (2024), Walters and Wilder (2023)	the author's documented search; gap statements for conceptual and design-science papers	four claims, two declared values, three declared definitions
Check	source check, both steps, for each source	method check of the search; warrant check, because absence is inferred from a search	Whetten's elements (Table 4); boundary conditions and values stated
Priority	P0: the opening argument fails without it	P0: the gap the paper addresses	P0 by definition: it is what the paper contributes
Tier	Supported	Emerging	Emerging
How it was set	two independent peer-reviewed sources agree	one screener and no search of Scopus, Web of Science or humanities indexes; the warrant check names the same gap	no higher than its weakest supporting claim (S2-2 and S3-4 are Emerging)
Qualifier in the text	reported findings ("found")	"I found none"	"I propose", "could", "should test"

Table 4: Whetten's elements applied to the contribution (C1).

Element	**Content**
What	A claim register; non-empirical papers; AI-assisted writing (each a declared definition).
How	Publishing the register exposes each step from grounds to claim to readers and reviewers.
Why	Citation checks pass weak reasoning (S1-7), reporting guidelines check reporting (S2-5), no checklist was found for the reasoning (S2-2), and each claim needs the check its grounds call for (S3-4).
Who, where, when	Papers whose contribution is an argument, and the interpretive parts of empirical papers; other writing, such as journalism, untested; frameworks from Western traditions.
What's new	The combination: each part is used elsewhere (S4-6 to S4-8, S3-6), and none of the compared approaches combines them for academic papers (S4-9).
Values	Published reasoning should be checkable (D1); a proposal should be tested before it is adopted (D7).
Objections	Peer review already judges arguments: a register would support review, not replace it. The register burdens authors: a shorter reviewer form is named as an option (Section 5).

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